

Verdant Coil: Game Elements & Mechanics Design Document

This document outlines all greenlit systems for *Verdant Coil*, including player stats, world elements, resources, tile types, level structure, and runtime mechanics.



Player Unit: Crawler Stats

Stat	Description
Health	Hitpoints; reduced by acid pools and other hazards.
Armor	Reduces or nullifies damage from specific tile effects.
Speed	Affects tile traversal rate (used if action delay is implemented).
Traction	Reduces slowing effects from sticky or slime surfaces.
Perception	Reveals hidden or camouflaged tiles like membrane traps.
Acid Reserve	Limits how many Digest Walls can be dissolved per run.
Upgrades	Binary toggles: Hardened Skin, Acid Sac, Photoreceptors, etc.



Game World Elements



Structural Tiles

Tile Type	Description
Flesh Tile	Walkable base tile.
Digest Wall	Destructible using Acid Sac.
Pulse Gate	Impassable unless matching upgrade is active.
Heartroot	Goal tile. Ends the level. Must be placed in every level.
Trigger Node	Sends signal to activate linked traps or toggles.
Membrane Tile	Hides what's beneath: hazards or rewards.
Camouflage Tile	Visually masks enemies or traps underneath.

AOE Hazard Tiles

Tile Type	Description
Acid Pool	Deals damage on contact.
Sticky Web	Slows movement unless countered by upgrade.
Quivering Spores	Emits AOE debuff or damage cloud when triggered.

Environmental Surfaces (*optional tags layered on tiles*)

Type	Effect
Slime	Reduces traction.
Water	Blocks certain upgrades or slows movement.
Soil	Supports growth-based traps or expansion modules.
Mycelium	May spawn spores or mutate tiles over time.
Acid	Permanent damage overlay.

Projectile Tiles

Tile Type	Description
Guard Nodule	Stationary turret that fires in a direction on interval or trigger.
Spine Shooter	Fires a straight-line projectile on a timing loop.
Tendrill Lash	Hits tiles in a predefined area or arc every few turns.

Special Functional Tiles

Tile Type	Description
Trigger Node	Activates linked tiles when stepped on.
Camouflage	Hides visible tile effects underneath.
Membrane	Masks acid, reward, or trap visuals until stepped on.

Resource System (Cross-Use)

Resource	Earned From	Used For
Nutrient	Completing other players' levels	Base Building: new modules, grid expansion

Resource	Earned From	Used For
Sporeprint	When your level is completed	Crawler Upgrades: unlock or enhance abilities

Level Structure & Layout

Grid Format

- Default: **24 x 24 tile grid**
- Expandable up to **48 x 48** by spending **Nutrient**
- Enforced **Biomass Budget** caps number of tiles placed

Start & Goal

- **Start:** Bottom-center of grid (default position)
- **Heartroot:** Can be placed anywhere; must be reachable via Flesh Tiles

Placement Rules

- One Heartroot per level
- At least one continuous path required (validation enforced)
- Levels must be completed by the creator before publishing

Runtime Mechanics: Turn-Based System

Movement & Action Flow (One Turn = One Player Move)

1. **Player moves one tile**
2. **AOE tiles resolve** (e.g. Acid, Spores)
3. **Trigger Nodes activate** linked tiles (e.g. trap doors, turrets)
4. **Projectiles move or fire** (e.g. Guard Nodule, Spine Shooter)
5. **Enemies** (if added) take actions last

Justification

- Deterministic; supports tactical depth
- Easier to debug, test, and simulate during development
- Compatible with grid-based builder mode

Sample Layout (Text Grid)





Legend:

- = Start (Crawler)
- = Heartroot (Goal)
- = Digest Wall
- = Guard Nodule
- = Trigger Node
- = Flesh Tiles / Walkable Path

This layout and mechanics system lays the foundation for a layered, asynchronous, and deeply strategic experience that encourages both creativity and problem-solving.